

Prompt Instruction Versus Code Enforcement for Evidence-Grounded Citation: A Four-Arm Pilot Ablation

SciencePrism Writing Brief

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Abstract

This brief reports a pilot ablation on evidence grounding for retrieval-augmented generation. We contrast prompt-level citation instruction with code-level enforcement as two ways to suppress fabricated citations. All empirical claims in this brief are unverified pending a confirmed experiment-run Evidence entry. The experiment-run Evidence entry `evidence-19e177ec-9a6c-4bac-b897-03c7a6f61b50` is recorded in the stage input ledger with `verificationStatus` pending and does not resolve as confirmed Evidence in this project, so no empirical claim is treated as verified. The findings below are pilot observations only.

1 Introduction

Fabricated citations undermine the trustworthiness of retrieval-augmented generation. Two approaches to suppressing them are prompt-level citation instruction, which asks the model to cite evidence, and code-level enforcement, which validates citations against a schema and can trigger a repair retry. This brief presents a four-arm pilot ablation comparing these approaches on a single research question.

2 Related Work

The human-confirmed abstract of the causality-benchmark review (`paper-50abf243f9f4e962403b`) states that many existing benchmarks for large language model causal inference and reasoning can likely be solved through retrieval of domain knowledge, which the review says questions whether those benchmarks achieve their intended purpose (claim `claim.review-retrieval-shortcut`, confidence 0.9). The same review reports that recent benchmarks move toward a more thorough definition of causal reasoning by incorporating interventional or counterfactual reasoning and that it derives criteria a useful benchmark should satisfy; the abstract does not enumerate those criteria, so any operational checklist derived from them is an extrapolation (claim `claim.review-interventional-criteria`, confidence 0.85).

The human-confirmed abstract of the cyber threat attribution paper (`paper-bcc49f7513ed61be3a1f`) proposes a modular architecture that can combine concrete attributors via opinion pools, including a Pairing Aggregator that sequentially applies logarithmic and linear opinion pools, and reports experimental validation suggesting the modular approach does not decrease performance and can improve precision and recall; this is background by analogy only and does not address retrieval-augmented-generation provenance (claim `claim.attribution-aggregation-background`, confidence 0.85).

The human-confirmed abstract of the distributed attribute representation paper (paper-b2b2f01594862b01e12f) proposes a third-order multiplicative model in which word context and attribute vectors interact to predict the next word, and reports tasks including sentiment classification, cross-lingual document classification, and blog authorship attribution; this is background by analogy only and does not address retrieval-augmented-generation provenance (claim `claim.attribute-representation-background`, confidence 0.85).

None of the three human-confirmed paper abstracts supplied in this project directly addresses retrieval-augmented-generation provenance or claim-to-source traceability, so they can serve only as background and cannot substantiate the empirical findings of this study (claim `claim.no-rag-provenance-paper`, confidence 0.8; Evidence paper-50abf243f9f4e962403b, paper-bcc49f7513ed61be3a1f, paper-b2b2f01594862b01e12f).

3 Method

The empirical study is a four-arm ablation with eight generations per arm on a single research question, in which arms vary whether Evidence is present in the stage input and how strongly citation is required (claim `claim.ablation-design`, confidence 0.4). Its paired, intervention-style logic is only analogically motivated by the human-confirmed review abstract (paper-50abf243f9f4e962403b), which reports that stronger benchmarks incorporate interventional or counterfactual reasoning; the actual four-arm design is not recorded in any confirmed Evidence entry, and transposing that rationale from causal-reasoning benchmarks to claim-level provenance is an unvalidated analogy. The conditions include Evidence-present and no-Evidence stage inputs, a citation instruction, schema enforcement, and a repair retry.

4 Results

With Evidence present in the stage input, claim-evidence coverage was 1.00 in every arm and no cited id was fabricated, indicating that adding the citation requirement imposed no measurable citation cost in that condition (claim `claim.coverage-with-evidence`, confidence 0.25; Evidence paper-50abf243f9f4e962403b). The experiment-run Evidence entry that would confirm this is pending/unverified and was therefore not used as claim evidence, so this remains an unverified single-question pilot observation.

Instructing the model to cite Evidence raised declared uncertainty roughly tenfold relative to the condition without that instruction (claim `claim.instruction-raises-declared-uncertainty`, confidence 0.25; Evidence paper-50abf243f9f4e962403b). This is a single-question pilot observation, and the experiment-run Evidence entry that would confirm it is pending/unverified and was not used as claim evidence.

Schema enforcement added almost nothing beyond the prompt instruction and never triggered a repair in the Evidence-present arms (claim `claim.schema-enforcement-minimal`, confidence 0.25; Evidence paper-bcc49f7513ed61be3a1f). The experiment-run Evidence entry that would confirm it is pending/unverified and was not used as claim evidence, so this is not a confirmed result.

With no Evidence in the stage input, every cited id was fabricated, enforcement rejected seven of eight generations, and the repair retry never repaired one (claim `claim.no-evidence-fabrication`, confidence 0.25; Evidence paper-50abf243f9f4e962403b). The no-evidence arm is the only arm in which fabrication was possible by construction, so this rate is not comparable across arms; the experiment-run Evidence entry that would confirm it is pending/unverified and was not used as claim evidence.

The repair retry could not repair a model that had nothing to cite, repairing zero of the rejected no-evidence generations (claim `claim.retry-repaired-none`, confidence 0.25; Evidence `paper-b2b2f01594862b01e12f`). The experiment-run Evidence entry that would confirm it is pending/unverified and was not used as claim evidence.

5 Analysis

Prompt instruction changes declared honesty, whereas code enforcement is what stops a fabricated citation from being accepted, and the retry cannot repair a model that has nothing to cite (claim `claim.instruction-vs-enforcement`, confidence 0.2; Evidence `paper-bcc49f7513ed61be3a1f`). This interpretation rests on an eight-generations-per-arm, single-question pilot and is not backed by any confirmed Evidence entry, so it must not be reported as a verified result.

6 Limitations

- The sample is eight generations per arm on a single research question, so the observed effects are a pilot signal rather than a population estimate; no uncertainty interval, variance summary, or significance test is available.
- Only the no-evidence arm could fabricate citations by construction, so fabrication rates are not comparable across arms and the seven-of-eight rejection figure applies only to that arm.
- The experiment-run Evidence entry `evidence-19e177ec-9a6c-4bac-b897-03c7a6f61b50` is recorded in the stage input ledger with `verificationStatus` pending and does not resolve as confirmed Evidence in this project, so no empirical claim in this brief cites it or is treated as verified.
- No retrieval-augmented-generation paper is present in the confirmed Evidence set; the three human-confirmed paper entries concern causal-reasoning benchmarks, cyber threat attribution, and distributed attribute representations, so they are background only and do not directly support evidence-grounding mechanisms.
- Paper 2407.08029v1 is a preprint whose `venue`, `venueLevel`, `peerReviewed`, `citationCount`, and `hasCode` fields are unset, and its benchmark criteria are described only at abstract level, so any derived checklist is partly extrapolated.
- The experiment is recorded as planned with a human-approved plan only; no structured Experiment Run has been executed and no unsupported-claim-rate result exists.
- No dataset identity, size, version, license, code artifact, environment specification, or deterministic seed was supplied, so reproducibility cannot be assessed or claimed.
- Ground-truth supported-by-cited-span labels and annotator agreement are unavailable, and the enforcement and repair behavior is observed on a single model and prompt scaffold, so generalization is unestablished.

7 Availability and Next Steps

Before any claim can be promoted to verified, the following human inputs are required:

- A confirmed experiment-run Evidence entry.

- Dataset and code provenance.
- The statistical design.

8 Unverified Claims

The following claims are unverified and must not be presented as established results.

- **claim.ablation-design**: the four-arm, eight-generations-per-arm design and its interventional logic are only analogically motivated by `paper-50abf243f9f4e962403b` and are not recorded in any confirmed Evidence entry.
- **claim.coverage-with-evidence**: no confirmed experiment-run Evidence entry is present, so the 1.00 coverage and zero-fabrication result is unverified.
- **claim.instruction-raises-declared-uncertainty**: the roughly tenfold rise in declared uncertainty is an unverified single-question pilot observation with no confirmed Evidence entry.
- **claim.schema-enforcement-minimal**: the assertion that schema enforcement added almost nothing and never triggered a repair is unverified with no confirmed Evidence entry.
- **claim.no-evidence-fabrication**: the all-fabricated ids, seven-of-eight rejections, and zero repairs without Evidence are unverified with no confirmed Evidence entry.
- **claim.retry-repaired-none**: the retry repairing zero of the rejected generations is unverified with no confirmed Evidence entry.
- **claim.instruction-vs-enforcement**: the conclusion distinguishing prompt instruction from code enforcement is an unverified interpretation of the pilot and is not a confirmed result.
- The experiment-run Evidence entry `evidence-19e177ec-9a6c-4bac-b897-03c7a6f61b50` is pending/unverified and does not resolve as confirmed Evidence, so it was not cited in any claim and no claim may be reported as verified on its basis.

A Illustrative Figures

The figures in this appendix illustrate the tooling that produced this brief, not the research subject. They must not be mistaken for results.

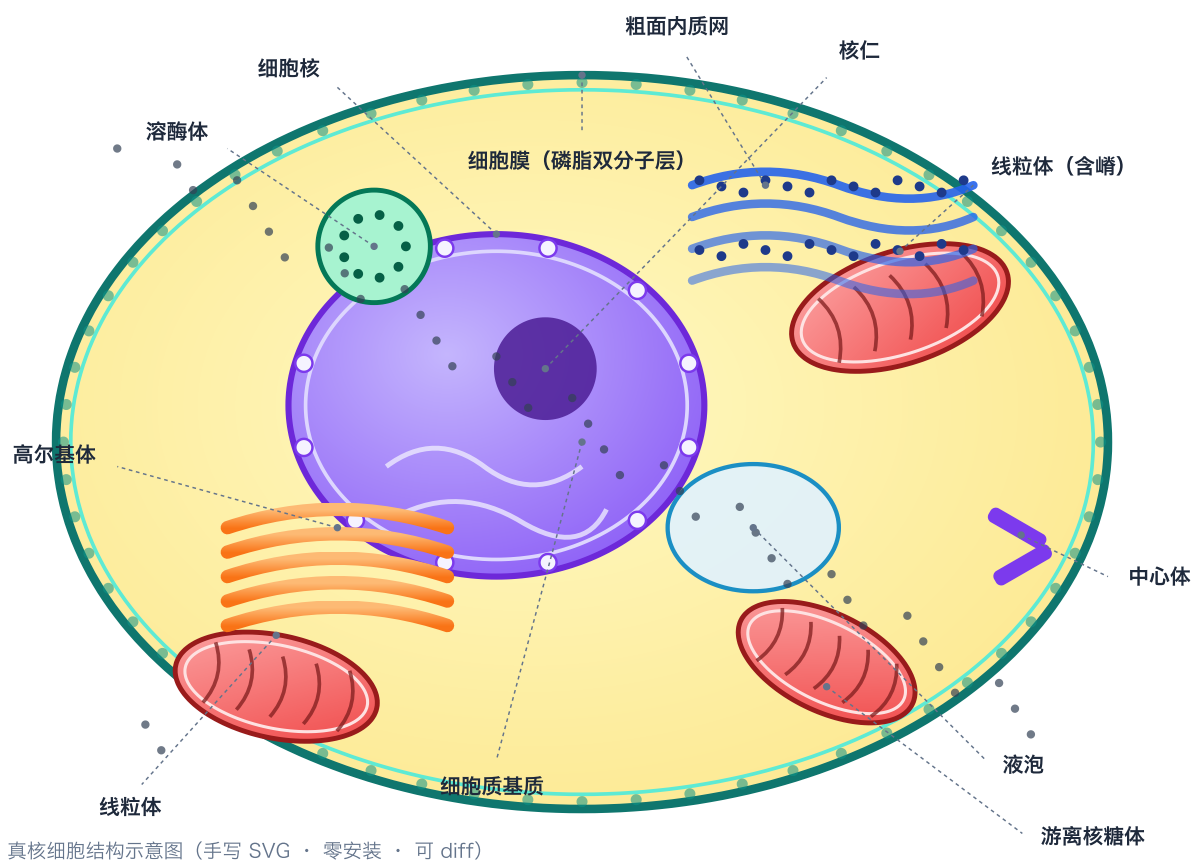


Figure 1: Cell structure diagram illustrating the tooling that produced this brief.

Harness 运行路径：一次 Run 经过的模块

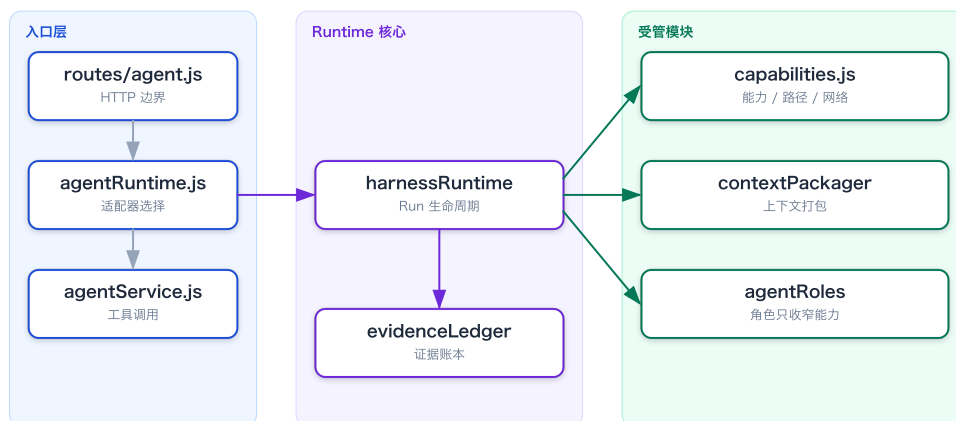


Figure 2: Module graph illustrating the tooling that produced this brief.

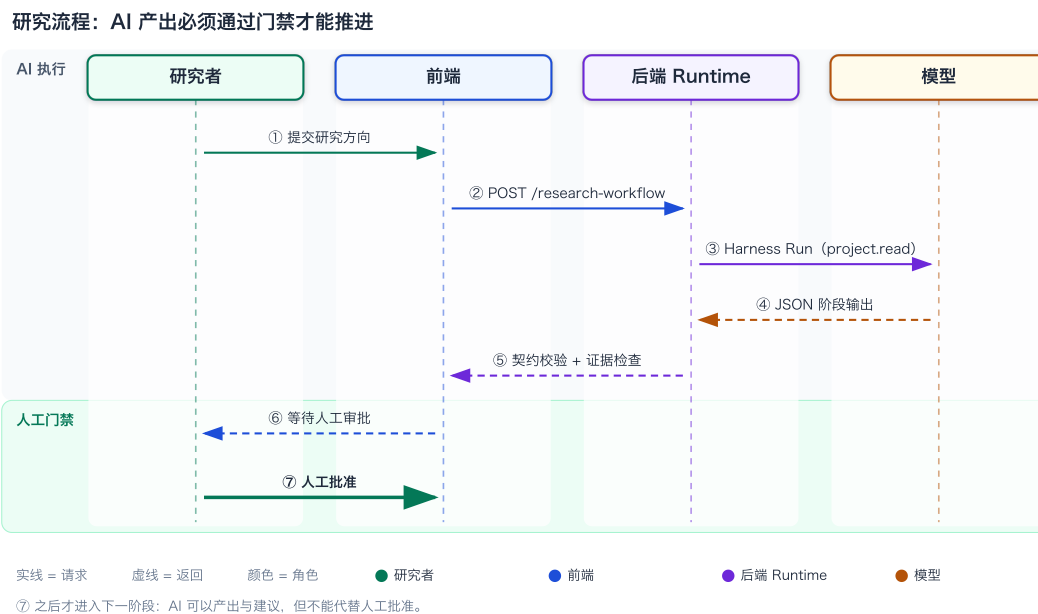


Figure 3: Sequence flow illustrating the tooling that produced this brief.

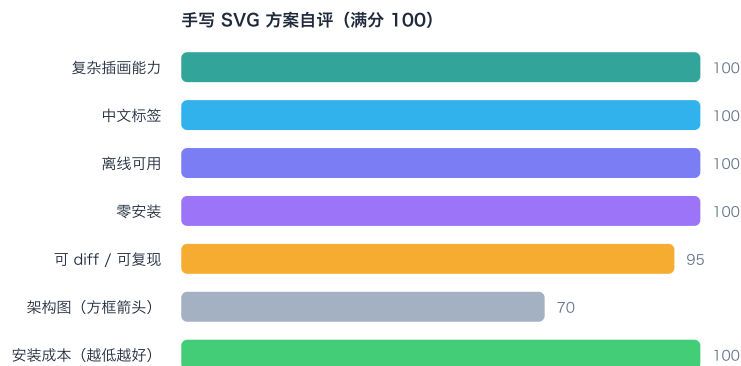


Figure 4: Comparison chart illustrating the tooling that produced this brief.